

Computational Finance

Lecture 8

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Lecture 8 – Jump Diffusion Processes

- Poisson processes
- Jump diffusion processes
- Affine jump diffusion processes

Jumps Are Observed in Stock Prices

- Below is the SPX plot from Jan-13-2020 till Aug-13-2021, covering the beginning part of the COVID-19 pandemic.



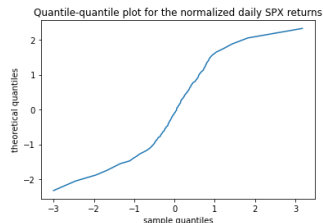
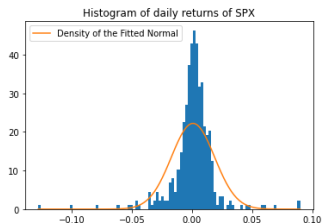
Fat-Tailed Distribution

- A **fat-tailed** distribution is a probability distribution that exhibits a large skewness or kurtosis, relative to that of either a normal distribution or an exponential distribution.
 - The skewness of a random variable X is the third standardized moment:

$$\mathbb{E}\left[\left(\frac{X - \mu}{\sigma}\right)^3\right]$$

- The kurtosis is the fourth standardized moment:

$$\mathbb{E}\left[\left(\frac{X - \mu}{\sigma}\right)^4\right]$$



Poisson Random Variable

- **Definition:** a discrete random variable X is said to have a Poisson distribution with parameter $\lambda > 0$ if its probability mass function is

$$f(k) = \mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k \in \mathbb{N}$$

- k is the number of occurrences;
 - $!$ is the factorial function (note $0! = 1$).
- Note that $\lambda = \mathbb{E}[X]$ because

$$\sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} \cdot k = \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-1)!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda.$$

- Further, the variance of X also equals λ .

Model Jump-Counts by Poisson Distributions

- Consider a fixed t representing the length of a time interval.
- The number of certain events (e.g., jumps in asset prices) occurring in that interval can be seen as a discrete random variable.
- We can model it using the Poisson distribution:

$$\mathbb{P}(X = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k \in \mathbb{N}$$

- λ : intensity rate.

Poisson Process (1/4)

- Generalizing the earlier idea: the Poisson process N_t is a **counting stochastic process** with an intensity rate λ , satisfying the following properties:
 - $N_{t=0} = 0$;
 - Has independent and stationary increments.
- The number of events in any interval of length t is a Poisson random variable with parameter λt , i.e.,

$$\mathbb{P}(N_t = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k \in \mathbb{N}.$$

Poisson Process (2/4)

- By definition, the Poisson process N_t with a constant intensity rate λ can be re-written as

$$N_t = \int_0^t dN_s = \sum_{\tau_k \leq t} dN(\tau_k),$$

where $\{\tau_k\}$ are the (random) time points of the jumps of the Poisson process.

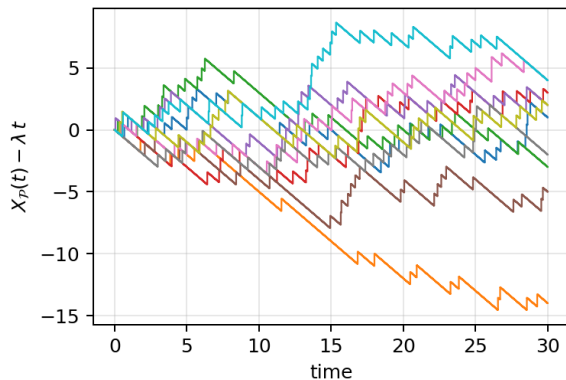
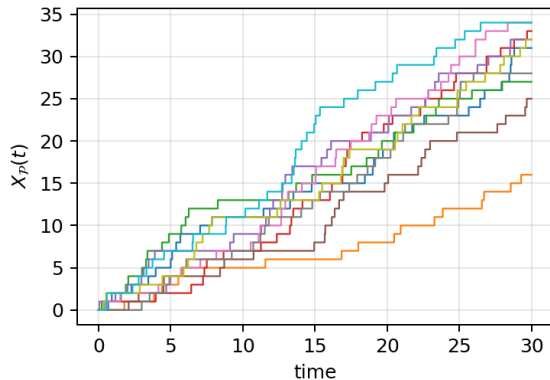
- $\int_0^t dN_s - \lambda t$ is called the compensated Poisson process.

- Note that the compensated Poisson process $\int_0^t dN_s - \lambda t$ is a martingale, because
 - $N_{t+\Delta t} - N_t$ is independent of $\{N_s, s \leq t\}$;
 - $N_{t+\Delta t} - N_t$ is stationary, i.e., $N_{t+\Delta t} - N_t \sim N_{\Delta t}$;
 - $\mathbb{E}[N_{t+\Delta t} - N_t] = \lambda \Delta t$;
 - $\mathbb{E}\left[\left|\int_0^t dN_s - \lambda t\right|\right] < +\infty$ for any t ;
 - the last inequality follows from

$$\mathbb{E}\left[\left|\int_0^t dN_s - \lambda t\right|\right] < \mathbb{E}\left[\int_0^t dN_s\right] + \lambda t = \mathbb{E}[N_t] + \lambda t = 2\lambda t.$$

Poisson Process (4/4)

Monte Carlo paths for the Poisson (left) and compensated Poisson (right), $\lambda = 1$



With Time-Dependent Intensity (1/2)

- The probability of jumps in asset prices is time varying $\Rightarrow$ the intensity rate λ can be generalized as a function of t , i.e., $\lambda(t)$,
- Such that the probability of k events occurring in the time interval $(a, b]$, with $N_{(a,b]} \equiv N_b - N_a$, is

$$\mathbb{P}(N_{(a,b]} = k) = \frac{\Lambda(a, b)^k e^{-\Lambda(a,b)}}{k!}, \quad k \in \mathbb{N},$$

with $\Lambda(a, b) = \int_a^b \lambda(s) ds$.

With Time-Dependent Intensity (2/2)

- Theoretically, it is proved that such an extension does exist.
- Correspondingly, N_t is called an **inhomogeneous Poisson process** satisfying the following four properties:
 - $N_0 = 0$;
 - Has independent increments;
 - $\mathbb{P}(N_{t+h} - N_t = 1) = \lambda(t) h + O(h^2)$;
 - $\mathbb{P}(N_{t+h} - N_t \geq 2) = O(h^2)$.
- Again, we can show that $\mathbb{E}[N_t] = \int_0^t \lambda(s) ds$, which further implies that

$$\int_0^t dN_s - \int_0^t \lambda(s) ds$$

is a martingale.

With Stochastic Intensity (1/2)

- Further, the intensity $\lambda(t)$ itself can be a stochastic process.
- A Poisson process with a stochastic intensity is called a **Cox process**, **compound Poisson process**, or a **doubly stochastic Poisson process**.
- Note that, conditional on the path of the stochastic process $\lambda(t)$, N_t is just a Poisson process with time-dependent intensity.

With Stochastic Intensity (2/2)

- Mathematically, given the filtration $\{\mathcal{F}_t\}$ to which both the Poisson process and the stochastic intensity process $\lambda(t)$ are adapted, and assuming $\int_0^T \lambda(t) dt < +\infty$ almost surely (a.s.) for any T , the probability that $N_{(a,b]} = k$ is defined as

$$\mathbb{P}(N_{(a,b]} = k \mid \mathcal{F}_a) = \mathbb{E} \left[\frac{\Lambda(a,b)^k e^{-\Lambda(a,b)}}{k!} \mid \mathcal{F}_a \right], \quad k \in \mathbb{N},$$

where $\Lambda(a,b) = \int_a^b \lambda(s) ds$.

- Again, the compensated Poisson process $\int_0^t dN_s - \int_0^t \lambda(s) ds$ is a martingale.

Paths of Poisson Processes (1/2)

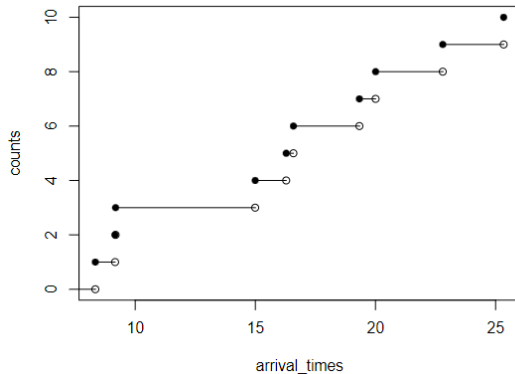
- Poisson processes are càdlàg, i.e., a path is everywhere right-continuous and with left limits.
- Roughly speaking, a function is right-continuous if no jump occurs when the limit point is approached from the right.
- That is, a jump (if occurring at t) is not predictable before t . Mathematically, N_t is adapted to $\mathcal{F}_t$ but not adapted to

$$\mathcal{F}_{t-} := \sigma(\mathcal{F}_{t-h}, h > 0)$$

(all the information available before t). Therefore, large price movements modelled by Poisson jumps cannot be predicted.

Paths of Poisson Processes (2/2)

- Visually, the path of a Poisson process looks like a step function, with left limits at discontinuities.



Jump Sizes (1/2)

- The sizes of jumps in asset prices are not constant either. Now we are ready to introduce **random jump sizes**. Consider

$$dY_t = J_t dN_t.$$

- N_t is the Poisson process that counts the number of jumps. There is a jump whenever $dN_t = N_t - N_{t-} = 1$. Note that $dN_t < 2$ because

$$\mathbb{P}(N_{t+dt} - N_t \geq 2) = O(dt^2).$$

- J_t denotes the jump size, assuming that J_t is a **left**-continuous process.

Jump Sizes (2/2)

- A fundamental condition required for modelling asset prices is the martingale property for the ratio of an asset price to the numeraire price.
- Recall that in the Black–Scholes model, the asset price divided by the bank-account numeraire is a martingale under the risk-neutral measure Q .
- We need the same martingale property in a jump-diffusion model. For that we need Itô's lemma for jumps to find out the drift term under Q .

Itô's Lemma for Jump Diffusions (1/2)

- Itô's lemma can be extended to semi-martingales, which include the jump-diffusion processes of our interest. It is important to note that the semi-martingales (including the jump diffusions) are càdlàg processes.
 - “In probability theory, a real-valued stochastic process X is called a semi-martingale if it can be decomposed as the sum of a local martingale and a càdlàg adapted finite-variation process. Semi-martingales are ‘good integrators’, forming the largest class of processes with respect to which the Itô integral and the Stratonovich integral can be defined. The class of semi-martingales is quite large (including, for example, all continuously differentiable processes, Brownian motion and Poisson processes).”
 - “A local martingale is a type of stochastic process satisfying the localized version of the martingale property. Every martingale is a local martingale; every bounded local martingale is a martingale.”

Itô's Lemma for Jump Diffusions (2/2)

- Consider the jump diffusion process

$$dY_t = \mu_t dt + \sigma_t dW_t + J_t dN_t.$$

- Itô's lemma needs to be adjusted to ensure the jumps are correctly added, i.e.,

$$\begin{aligned} dg(Y_t, t) &= \frac{\partial g}{\partial t}(Y_t, t) dt + \frac{\partial g}{\partial Y}(Y_t, t) dY_t^c + \frac{1}{2} \frac{\partial^2 g}{\partial Y^2}(Y_t, t) dY_t^c dY_t^c \\ &\quad + [g(Y_t, t) - g(Y_{t-}, t)] dN_t \\ &= \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial Y} (\mu_t dt + \sigma_t dW_t) + \frac{1}{2} \frac{\partial^2 g}{\partial Y^2} \sigma_t^2 dt \\ &\quad + [g(Y_{t-} + J_t, t) - g(Y_{t-}, t)] dN_t \end{aligned}$$

- g is twice differentiable w.r.t. Y and differentiable w.r.t. t ;
- $Y_{t-} := \lim_{s \rightarrow t, s < t} Y_s$, the left limit of Y_t ;
- Y_t^c is the continuous part of Y_t (after excluding all the discontinuous jumps):
 $Y_t^c = Y_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s$;
- equivalently, $Y_t^c = Y_t - \sum_{0 \leq s \leq t} (Y_s - Y_{s-})$, with s being the jump time prior to t .

Itô's Product Rule for Jump Diffusions (1/2)

- Consider two jump-diffusion processes Y_t and Z_t ,

$$dY_t = \mu_t^Y dt + \sigma_t^Y dW_t^Y + J_t^Y dN_t^Y,$$

$$dZ_t = \mu_t^Z dt + \sigma_t^Z dW_t^Z + J_t^Z dN_t^Z.$$

- The Itô product rule is

$$d(Y_t Z_t) = Y_{t-} dZ_t + Z_{t-} dY_t + d[Y, Z]_t,$$

- $[Y, Z]_t$ is the **covariation** of Y and Z :

$$[Y, Z]_t := \lim_{\max |t_k - t_{k-1}| \rightarrow 0} \sum_{k=1}^n (Y_{t_k} - Y_{t_{k-1}})(Z_{t_k} - Z_{t_{k-1}}) = \int_0^t \sigma_s^Y \sigma_s^Z \rho ds + \sum_{s \leq t} J_s^Y J_s^Z.$$

- Note that ρ is the correlation between the Brownian motions W^Y and W^Z , and $\sum_{s \leq t} J_s^Y J_s^Z$ measures the jumps of Y and Z occurring at the same time.

Itô's Product Rule for Jump Diffusions (2/2)

- For $Y = Z$, $[Y, Y]_t$ is the quadratic variation of Y :

$$[Y, Y]_t := \lim_{\max |t_k - t_{k-1}| \rightarrow 0, t_0=0, t_K=t} \sum (Y_{t_k} - Y_{t_{k-1}})^2 = \int_0^t (\sigma_s^Y)^2 ds + \sum_{s \leq t} (J_s^Y)^2.$$

- For simplicity, we often write $d[Y, Z]_t$ as $dY_t dZ_t$. This is consistent with our earlier notation since, without jumps,

$$dY_t dZ_t = \sigma_t^Y \sigma_t^Z dW_t^Y dW_t^Z = \sigma_t^Y \sigma_t^Z \rho dt.$$

A Jump-Diffusion Model for Stock Prices (1/3)

- Consider the stock price evolving as

$$dS_t = \mu S_{t-} dt + \sigma S_{t-} dW_t + J_t S_{t-} dN_t.$$

- For simplicity we let J_t be a continuous random variable independent of S_t and N_t .
- The question is: what is μ under the risk-neutral measure Q such that S_t/M_t is a martingale?

A Jump-Diffusion Model for Stock Prices (2/3)

- Apply the Itô product rule

$$d(Y_t Z_t) = Y_{t-} dZ_t + Z_{t-} dY_t + d[Y, Z]_t$$

to $S_t/M_t = e^{-rt} S_t$, we have

$$\begin{aligned} d(S_t/M_t) &= -re^{-rt} S_{t-} dt + e^{-rt} dS_t + de^{-rt} dS_t \\ &= e^{-rt} [-r S_{t-} dt + dS_t] + \cancel{de^{-rt} dS_t} \\ &= e^{-rt} [(\mu - r) S_{t-} dt + \sigma S_{t-} dW_t + J_t S_{t-} dN_t]. \end{aligned}$$

- We know that $\sigma S_{t-} dW_t$ is a martingale.

A Jump-Diffusion Model for Stock Prices (3/3)

- Hence, if $\mu_t = r - \lambda_t J_t$, then $\mathbb{E}^Q[d(S_t/M_t)] = 0$, which makes S_t/M_t a martingale:

$$\begin{aligned}\mathbb{E}^Q[d(S_t/M_t)] &= e^{-rt} \mathbb{E}^Q[(\mu_t - r) S_{t-} dt + \sigma S_{t-} dW_t + J_t S_{t-} dN_t] \\ &= e^{-rt} S_{t-} \mathbb{E}^Q[-\lambda_t J_t dt + J_t dN_t] = 0.\end{aligned}$$

- If J_t is independent of λ_t , then alternatively we can set $\mu_t = r - \lambda_t \mathbb{E}^Q[J_t]$ so that $\mathbb{E}^Q[d(S_t/M_t)] = 0$. In this case, the stock price under Q is

$$dS_t = (r - \lambda_t \mathbb{E}^Q[J_t]) S_{t-} dt + \sigma S_{t-} dW_t + J_t S_{t-} dN_t.$$

SDE of Log-Price in a Jump-Diffusion Model

- Apply Itô's lemma:

$$\begin{aligned}d \ln(S_t) &= \frac{1}{S_{t-}} dS_t^c - \frac{1}{2S_{t-}^2} dS_t^c dS_t^c + [\ln S_t - \ln S_{t-}] dN_t \\&= \frac{1}{S_{t-}} (\mu S_{t-} dt + \sigma S_{t-} dW_t) - \frac{1}{2S_{t-}^2} \sigma^2 S_{t-}^2 dt + \ln(1 + J_t) dN_t \\&= (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t + \ln(1 + J_t) dN_t.\end{aligned}$$

- Above we use $[\ln S_t - \ln S_{t-}] dN_t = \ln(1 + J_t) dN_t$. To see it, note that $\Delta S_t := S_t - S_{t-} = J_t S_{t-} dN_t$, which implies $S_t = S_{t-}(1 + J_t)$ whenever there is a jump $dN_t = 1$. Equivalently, $\ln S_t - \ln S_{t-} = \ln(1 + J_t)$ at the time of a jump.
- Plugging in $\mu = r - \lambda_t \mathbb{E}^Q[J_t]$ yields

$$d \ln(S_t) = (r - \lambda_t \mathbb{E}^Q[J_t] - \frac{1}{2}\sigma^2) dt + \sigma dW_t + \ln(1 + J_t) dN_t.$$

- We want to derive the PDE for the price $V(S_t, t)$ of a payoff at maturity T under a jump-diffusion model.
- Again, we resort to the fact that V_t/M_t is a martingale under the risk-neutral measure Q .

- Apply Itô's lemma:

$$\begin{aligned}
 d(e^{-rt} V_t) &= -re^{-rt} V_{t-} dt + e^{-rt} dV_t + de^{-rt} dV_t \\
 &= -re^{-rt} V_{t-} dt + e^{-rt} \left[\frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS_t^c + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} dS_t^c dS_t^c + (V(S_t, t) - V(S_{t-}, t)) dN_t \right] \\
 &= e^{-rt} \left[-rV_{t-} dt + \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} \mu S_{t-} dt + \frac{\partial V}{\partial S} \sigma S_{t-} dW_t \right. \\
 &\quad \left. + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S_{t-}^2 dt + (V(S_{t-}(1+J_t), t) - V(S_{t-}, t)) dN_t \right] \\
 &= e^{-rt} \left[-rV_{t-} + \frac{\partial V}{\partial t} + \frac{\partial V}{\partial S} \mu S_{t-} + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S_{t-}^2 \right] dt \\
 &\quad + e^{-rt} \frac{\partial V}{\partial S} \sigma S_{t-} dW_t + e^{-rt} (V(S_{t-}(1+J_t), t) - V(S_{t-}, t)) dN_t.
 \end{aligned}$$

- We need $\mathbb{E}^Q[d(e^{-rt}V_t)] = 0$ such that V_t/M_t is a martingale.
- Note that the expectation of the jump of V is

$$\mathbb{E}^Q[(V(S_{t-}(1+J_t), t) - V(S_{t-}, t)) dN_t] = \mathbb{E}^Q[(V(S_{t-}(1+\mathbb{E}^Q[J_t]), t) - V(S_{t-}, t))] dN_t,$$

the equality is due to the assumption of independence made for J_t .

- Together with $\mu = r - \lambda_t \mathbb{E}^Q[J_t]$, we can tell $\mathbb{E}^Q[d(e^{-rt}V_t)] = 0$ if

$$-rV_{t-} + \frac{\partial V}{\partial t} + \frac{\partial V}{\partial S}(r - \lambda_t \mathbb{E}^Q[J_t])S_{t-} + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S_{t-}^2 + \mathbb{E}^Q[V(S_{t-}(1+\mathbb{E}^Q[J_t]), t) - V(S_{t-}, t)] dN_t = 0.$$

Two Popular Jump-Diffusion Models

- **Classical Merton's model:** the intensity λ is a constant and the jump size J_t is normally distributed, with mean μ_J and standard deviation σ_J . The density function reads

$$\nu_J(x) = \frac{1}{\sigma_J \sqrt{2\pi}} \exp\left(-\frac{(x - \mu_J)^2}{2\sigma_J^2}\right).$$

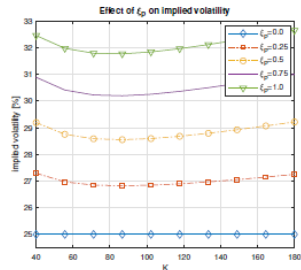
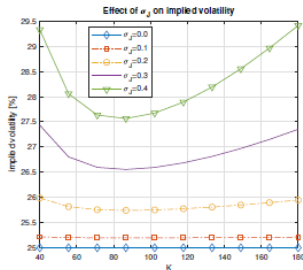
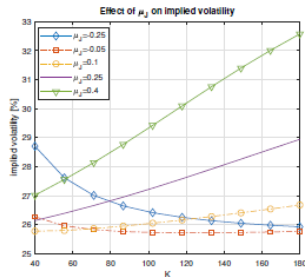
- **Non-symmetric double exponential (Kou's model):**

$$\nu_X(x) = p \eta_1 e^{-\eta_1 x} \mathbf{1}_{\{x \geq 0\}} + q \eta_2 e^{\eta_2 x} \mathbf{1}_{\{x < 0\}},$$

- where p and q are positive real numbers satisfying $p + q = 1$. To be able to integrate x over the real line it is required to have $\eta_1 > 1$ and $\eta_2 > 0$.

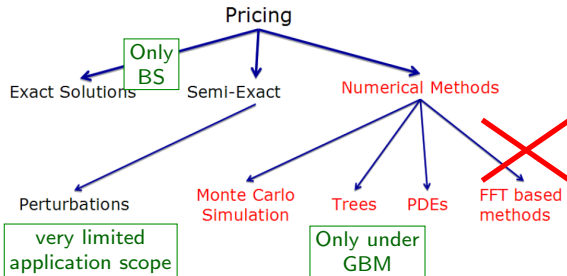
Implied Volatility

- **Merton's model:** the jump size J_t is normally distributed, with mean μ_J and standard deviation σ_J .
- σ_J has a significant impact on the curvature, λ (ξ_p in the plot) controls the overall level of the implied volatility, whereas μ_J influences the implied volatility slope (the skewness).



Pricing Techniques

There are several different approaches for obtaining information on the prices of options:



Should be **Fourier methods**;
The COS method is applicable
as long as ch.f. is known

Characteristic Function – Needed for Fourier Methods

- **Definition** – Fourier transform of the density function:

$$\psi^X(u, X_t, t, T) = \mathbb{E} \left[e^{i\omega \cdot X_T} \mid \mathcal{F}_t \right].$$

- **Example:**

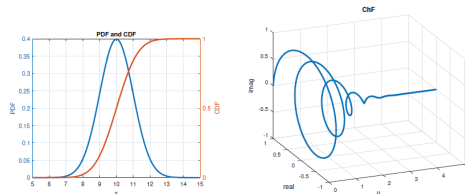


Figure: The CDF, PDF and characteristic function for an $\mathcal{N}(10, 1)$ random variable.

The CDF, PDF and characteristic function for an $\mathcal{N}(10, 1)$ random variable.

Affine Processes

- *Affine processes*: a class of time-homogeneous Markov processes that has arisen from a large and growing range of useful applications in finance.

Definition

The Markov process $(X, (\mathbb{P}_x)_{x \in D})$, and (P_t) , is called *affine* if, for every $t \in \mathbb{R}_+$, the characteristic function of $p_t(x, \cdot)$ has exponential-affine dependence on x . That is, if for every $(t, u) \in \mathbb{R}_+ \times \partial\mathcal{U}$ there exist $\phi(t, u) \in \mathbb{C}$ and $\psi(t, u) = (\psi^{\mathcal{Y}}(t, u), \psi^{\mathcal{Z}}(t, u)) \in \mathbb{C}^m \times \mathbb{C}^n$ such that

$$\begin{aligned} P_t f_u(x) &= e^{\phi(t, u) + \langle \psi(t, u), x \rangle} \\ &= e^{\phi(t, u) + \langle \psi^{\mathcal{Y}}(t, u), y \rangle + \langle \psi^{\mathcal{Z}}(t, u), z \rangle}, \quad \forall x = (y, z) \in D. \end{aligned} \tag{2.2}$$

- They are easy to solve since they allow tractable formulas of the [Fourier transform](#).
- We follow the notation of the paper presenting the transform analysis for the affine jump diffusions (Duffie–Pan–Singleton, 2000).

Affine Jump Diffusion (AJD) Models

- The SDE of an n -dimensional affine jump-diffusion process (AJD) can be described as

$$d\mathbf{X}_t = \mu(\mathbf{X}_t) dt + \sigma(\mathbf{X}_t) d\mathbf{W}_t + d\mathbf{Z}_t,$$

where

- Affine condition:** $\mu(\mathbf{x}) = \mathbf{K}_0 + \mathbf{K}_1 \mathbf{x}$, with $\mathbf{K}_0 \in \mathbb{R}^n$, $\mathbf{K}_1 \in \mathbb{R}^{n \times n}$;
- $\mathbf{W}_t$ is an n -dimensional Brownian motion;
- Affine condition:** $[\sigma(\mathbf{x})\sigma(\mathbf{x})^T]_{ij} = (\mathbf{H}_0)_{ij} + (\mathbf{H}_1)_{ij} \mathbf{x}$, with $\mathbf{H}_0 \in \mathbb{R}^{n \times n}$, $\mathbf{H}_1 \in \mathbb{R}^{n \times n \times n}$;
- $d\mathbf{Z}_t = \mathbf{J} dN_t$ is a jump process where the intensity is $\lambda(\mathbf{X}_t)$, and the jump sizes $\mathbf{J}$ are independent of the process $\mathbf{X}_t$ and have probability distribution ν ;
- $\lambda(\mathbf{x}) = l_0 + \mathbf{l}_1 \cdot \mathbf{x}$, with $l_0 \in \mathbb{R}$, $\mathbf{l}_1 \in \mathbb{R}^n$.

Fourier Transform Under AJD (1/3)

- Consider the transform $\psi^{\mathbf{X}}(\mathbf{u}, \mathbf{X}_t, t, T)$ for a complex vector $\mathbf{u}$ in $\mathbb{C}^n$:

$$\psi^{\mathbf{X}}(\mathbf{u}, \mathbf{X}_t, t, T) = \mathbb{E} \left[\exp \left(- \int_t^T R(\mathbf{X}_s) ds \right) e^{\mathbf{u} \cdot \mathbf{X}_T} \middle| \mathcal{F}_t \right].$$

- Moreover, the short rate is also an affine function of $\mathbf{X}_t$, i.e., $R(\mathbf{X}_t) = \rho_0 + \boldsymbol{\rho}_1 \cdot \mathbf{X}_t$ where $\rho_0 \in \mathbb{R}$, $\boldsymbol{\rho}_1 \in \mathbb{R}^n$. Note that the dot operation in $\mathbf{l}_1 \cdot \mathbf{x}$ and $\boldsymbol{\rho}_1 \cdot \mathbf{X}_t$ means the inner product of vectors for $n > 1$, or the scalar product for $n = 1$.

Fourier Transform Under AJD (2/3)

- A **key result** of AJD is that, under certain technical regularity conditions, it holds that

$$\psi^X(\mathbf{u}, \mathbf{x}, t, T) = \mathbb{E} \left[\exp \left(- \int_t^T R(\mathbf{X}_s) ds \right) e^{\mathbf{u} \cdot \mathbf{X}_T} \middle| \mathcal{F}_t \right] = e^{\alpha(\mathbf{u}, t, T) + \beta(\mathbf{u}, t, T) \cdot \mathbf{x}},$$

- α (scalar) and β (vector) satisfy the complex-valued ordinary differential equations (ODEs)

$$\begin{aligned} \dot{\beta}(\mathbf{u}, t, T) &= \rho_1 - \mathbf{K}_1^T \beta(t) - \frac{1}{2} \beta^T(t) \mathbf{H}_1 \beta(t) - l_1(\theta(\beta(t)) - 1), \\ \dot{\alpha}(\mathbf{u}, t, T) &= \rho_0 - \mathbf{K}_0 \cdot \beta(t) - \frac{1}{2} \beta^T(t) \mathbf{H}_0 \beta(t) - l_0(\theta(\beta(t)) - 1), \end{aligned}$$

- with boundary conditions $\beta(\mathbf{u}, T, T) = \mathbf{u}$ and $\alpha(\mathbf{u}, T, T) = 0$.

- The random jump size is characterized by its characteristic function

$$\theta(\mathbf{c}) = \int_{\mathbb{R}^n} \exp(\mathbf{c} \cdot \mathbf{z}) d\nu(\mathbf{z}).$$

- Note that $\beta^T(t) \mathbf{H}_1 \beta(t)$ denotes an n -dimensional vector with the k -th element being

$$\sum_{ij} \beta_i(t) (H_1)_{ijk} \beta_j(t).$$

An Example: Black–Scholes Case (1/3)

- With a substitution, $X_t = \ln(S_t)$, we have

$$U(X_t, t) = V(S_t, t),$$

$$\frac{\partial U}{\partial t} + r \frac{\partial U}{\partial X} + \frac{1}{2} \sigma^2 \left(-\frac{\partial U}{\partial X} + \frac{\partial^2 U}{\partial X^2} \right) - rU = 0.$$

- By setting $\tau = T - t$, we have

$$-\frac{\partial U}{\partial \tau} + \left(r - \frac{1}{2} \sigma^2 \right) \frac{\partial U}{\partial X} + \frac{1}{2} \sigma^2 \frac{\partial^2 U}{\partial X^2} - rU = 0.$$

- By the results of Duffie–Pan–Singleton, we probe the following solution:

$$U(X, t) := \phi(u, t) = \exp(\alpha(u, \tau) + \beta(u, \tau) X),$$

with boundary condition $\phi(u, t) = \exp(iuX)$.

An Example: Black–Scholes Case (2/3)

- By partial differentiation we have

$$\frac{\partial \phi}{\partial \tau} = \phi \cdot \left(\frac{\partial \alpha}{\partial \tau} + X \frac{\partial \beta}{\partial \tau} \right), \quad \frac{\partial \phi}{\partial X} = \phi \cdot \beta, \quad \frac{\partial^2 \phi}{\partial X^2} = \phi \cdot \beta^2.$$

- By substituting these quantities in the pricing PDE we have

$$-\left(\frac{\partial \alpha}{\partial \tau} + X \frac{\partial \beta}{\partial \tau} \right) + \left(r - \frac{1}{2} \sigma^2 \right) \beta + \frac{1}{2} \sigma^2 \beta^2 - r = 0.$$

From the above, we obtain the set of ODEs as follows:

$$\frac{\partial \beta}{\partial \tau} = 0, \quad \frac{\partial \alpha}{\partial \tau} = \left(r - \frac{1}{2} \sigma^2 \right) \beta + \frac{1}{2} \sigma^2 \beta^2 - r.$$

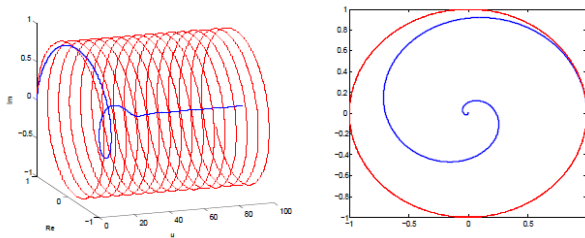
An Example: Black–Scholes Case (3/3)

- By using the boundary conditions we find

$$\beta(u, \tau) = iu, \quad \alpha(u, \tau) = \left(r - \frac{1}{2}\sigma^2\right)iu\tau - \frac{1}{2}\sigma^2u^2\tau - r\tau.$$

- So, the obtained solution (ch.f. of log-asset price under GBM) is

$$\phi(u, \tau) = \exp\left(\left(r - \frac{1}{2}\sigma^2\right)iu\tau - \frac{1}{2}\sigma^2u^2\tau - r\tau + iuX\right).$$



- Darrell Duffie, Jun Pan, Kenneth Singleton, *Transform Analysis and Asset Pricing for Affine Jump-Diffusions*, *Econometrica* 2000: 68(6).
<https://doi.org/10.1111/1468-0262.00164>.
- Steven E. Shreve, *Stochastic Calculus for Finance II, Continuous-Time Models*, Springer Finance Textbook.
- Ioannis Karatzas and Steven E. Shreve, *Brownian Motion and Stochastic Calculus*, Second edition, Springer Textbook.
- S. G. Kou, *A Jump-Diffusion Model for Option Pricing*, *Management Science*, Vol. 48, No. 8 (Aug., 2002), pp. 1086–1101.

- Proof of the key results of Duffie–Pan–Singleton (2000).

Proof (1/4)

- Consider

$$\varphi_t \equiv \varphi(\mathbf{X}_t, t) = \exp\left(-\int_0^t R(\mathbf{X}_s) ds\right) e^{\alpha(t)+\boldsymbol{\beta}(t)\cdot\mathbf{X}_t}.$$

- It is enough to show that φ_t is a martingale, for then $\varphi_t = \mathbb{E}[\varphi_T \mid \mathcal{F}_t]$, i.e.,

$$\exp\left(-\int_0^t R(\mathbf{X}_s) ds\right) e^{\alpha(t)+\boldsymbol{\beta}(t)\cdot\mathbf{X}_t} = \mathbb{E}_t\left[\exp\left(-\int_0^T R(\mathbf{X}_s) ds\right) e^{\alpha(t)+\boldsymbol{\beta}(t)\cdot\mathbf{X}_T}\right].$$

Multiplying both sides by $\exp\left(\int_0^t R(\mathbf{X}_s) ds\right)$ yields

$$e^{\alpha(t)+\boldsymbol{\beta}(t)\cdot\mathbf{X}_t} = \mathbb{E}_t\left[\exp\left(-\int_t^T R(\mathbf{X}_s) ds\right) e^{\alpha(t)+\boldsymbol{\beta}(t)\cdot\mathbf{X}_T}\right].$$

- Apply Itô's lemma:

$$\begin{aligned} d\varphi_t = & \frac{\partial \varphi}{\partial t} dt + \mu(\mathbf{X}_t) \frac{\partial \varphi}{\partial X} dt + \frac{\sigma(\mathbf{X}_t) \sigma(\mathbf{X}_t)^T}{2} \frac{\partial^2 \varphi}{\partial X^2} dt \\ & + \sigma(\mathbf{X}_t) \frac{\partial \varphi}{\partial X} dW_t + [\varphi_t(\mathbf{X}_{t-} + \mathbf{J}) - \varphi_t(\mathbf{X}_{t-})] dN_t. \end{aligned}$$

- The time- t expectation of $[\varphi_t(\mathbf{X}_{t-} + \mathbf{J}) - \varphi_t(\mathbf{X}_{t-})] dN_t$ is

$$\begin{aligned} \mathbb{E}_t[(\varphi_t(\mathbf{X}_{t-} + \mathbf{J}) - \varphi_t(\mathbf{X}_{t-})) dN_t] &= \int_{\mathbb{R}^n} (\varphi_t(\mathbf{X}_{t-} + \mathbf{y}) - \varphi_t(\mathbf{X}_{t-})) d\nu(\mathbf{y}) dN_t \\ &= \int_{\mathbb{R}^n} \varphi_{t-} \cdot (e^{\boldsymbol{\beta}(t) \cdot \mathbf{y}} - 1) d\nu(\mathbf{y}) dN_t \\ &= \varphi_t(\theta(\boldsymbol{\beta}(t)) - 1) \lambda(\mathbf{X}_t) dt. \end{aligned}$$

Proof (3/4)

- So φ_t is a martingale if

$$\frac{\partial \varphi}{\partial t} + \mu(\mathbf{X}_t) \frac{\partial \varphi}{\partial X} + \frac{\sigma(\mathbf{X}_t)\sigma(\mathbf{X}_t)^T}{2} \frac{\partial^2 \varphi}{\partial X^2} + \varphi_t(\theta(\beta(t)) - 1) \lambda(\mathbf{X}_t) = 0.$$

- Since

- $\frac{\partial \varphi}{\partial t} = [-R(\mathbf{X}_t) + \dot{\alpha}(t) + \dot{\beta}(t) \cdot \mathbf{X}_t] \varphi;$
- $\frac{\partial \varphi}{\partial X} = \beta(t) \varphi;$
- $\frac{\partial^2 \varphi}{\partial X^2} = \beta(t) \beta(t)^T \varphi,$

- the martingale condition requires

$$[-R(\mathbf{X}_t) + \dot{\alpha}(t) + \dot{\beta}(t) \cdot \mathbf{X}_t] + \mu(\mathbf{X}_t) \beta(t) + \beta(t) \frac{\sigma(\mathbf{X}_t)\sigma(\mathbf{X}_t)^T}{2} \beta(t)^T + (\theta(\beta(t)) - 1) \lambda(\mathbf{X}_t) = 0.$$

- Plugging in the affine functions $\mu(\mathbf{x})$, $\sigma(\mathbf{x})\sigma(\mathbf{x})^T$, $R(\mathbf{x})$, and $\lambda(\mathbf{x})$,

$$\begin{aligned} & -(\rho_0 + \rho_1 \cdot X_t) + \dot{\alpha}(t) + \dot{\beta}(t) \cdot X_t + (K_0 + K_1 \cdot X_t) \beta(t) \\ & + \frac{1}{2} \beta(t) (H_0 + H_1 \cdot X_t) \beta(t)^T + (\theta(\beta(t)) - 1)(l_0 + l_1 \cdot X_t) = 0, \end{aligned}$$

which leads to

$$\begin{aligned} & \left[-\dot{\alpha}(t) + \rho_0 - K_0 \cdot \beta(t) - \frac{1}{2} \beta(t)^T H_0 \beta(t) - l_0 (\theta(\beta(t)) - 1) \right] \\ & + \left[-\dot{\beta}(t) + \rho_1 - K_1^T \beta(t) - \frac{1}{2} \beta(t)^T H_1 \beta(t) - l_1 (\theta(\beta(t)) - 1) \right] X_t = 0. \end{aligned}$$

The ODEs ensure that the martingale condition holds. □